

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think> To determine the total time taken to download the 200GB file, we need to calculate the time taken for two separate parts: the first part where Carla downloads 60% of the file at a speed of 2 GB / minute, and the second part where she downloads the remaining 40% of the file after the restart. First, we calculate the time taken to download 60% of the 200GB file: 60% of 200 GB = 0.6 * 200GB = 120GB. At a speed of 2 GB / minute, the time taken for this part is 120GB / 2 GB / minute = 60 minutes. Then, we add the 20-minute restart time. Finally, we add the time taken to download the remaining 40% of the file: 40GB / 2 GB / minute = 20 minutes. So, the total time taken is 60 minutes + 20 minutes + 20 minutes = 100 minutes. </think><answer> 100</answer>

